



TECNOLÓGICO NACIONAL DE MÉXICO



Práctica tres: Sistema Cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

Table of Contents

Información general.....	1
Datos generales de la simulacion.....	1
Rendimiento del controlador Hipotenso	2
Hipertenso.....	2
Respuesta la función sinusoidal.....	2
Hipotenso.....	3
Hipertenso.....	4
Función: Respuesta a las señales.....	4
Función: Tratamiento (Hipotenso).....	5
Función: Tratamiento (Hipertenso).....	5

Información general



Nombre del alumno: **Mauricio Jesús Meraz Galeana**

Número de control: **18210139**

Correo institucional: **mauricio.meraz18@tectijuana.edu.mx**

Asignatura: **Modelado de Sistemas Fisiológicos**

Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

Datos generales de la simulacion

```
clc; clear all; close all;
```

```
tend = '10';
file = 'SistemaP3';
open_system(file);
parametrer.StopTime = tend;
parameters.Solver = 'ode15s';
```

Rendimiento del controlador Hipotenso

kP = 0.0003333

KI = 413.332

kD = 0

Rise time = 0.0054 s

settling time = 0.0097 s

Overshoot = 0 %

Peak = 1 V

Hipertenso

kP = 0.00022321

KI = 414.2852

kD = 0

Rise time = 0.0054 s

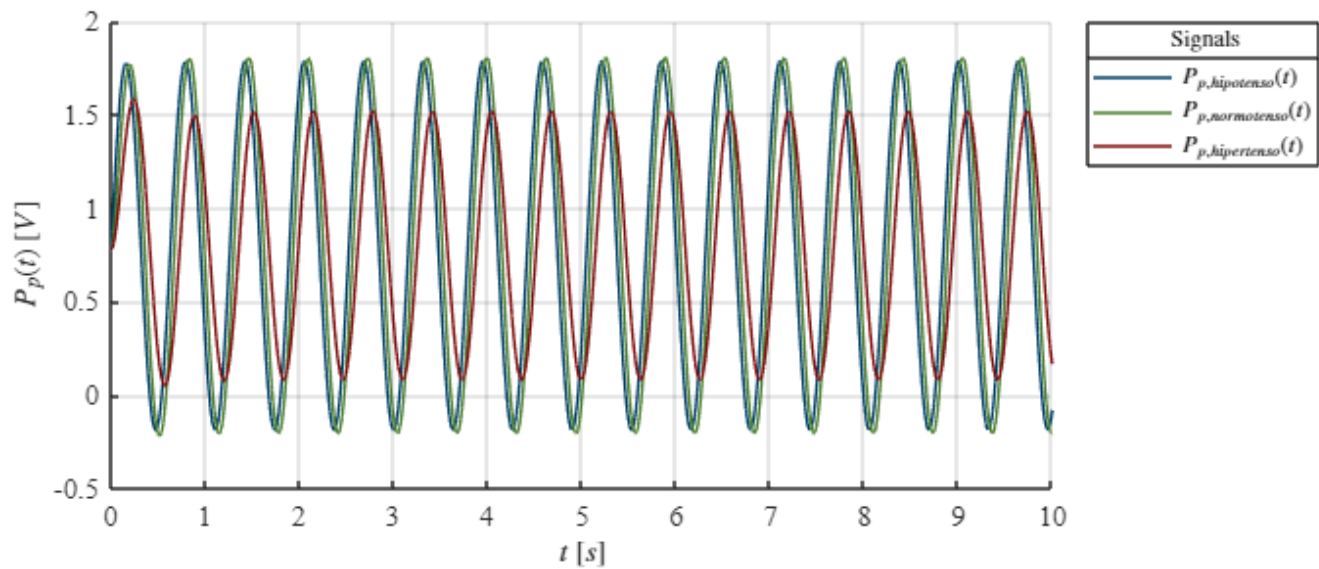
settling time = 0.0097 s

Overshoot = 0 %

Peak = 1 V

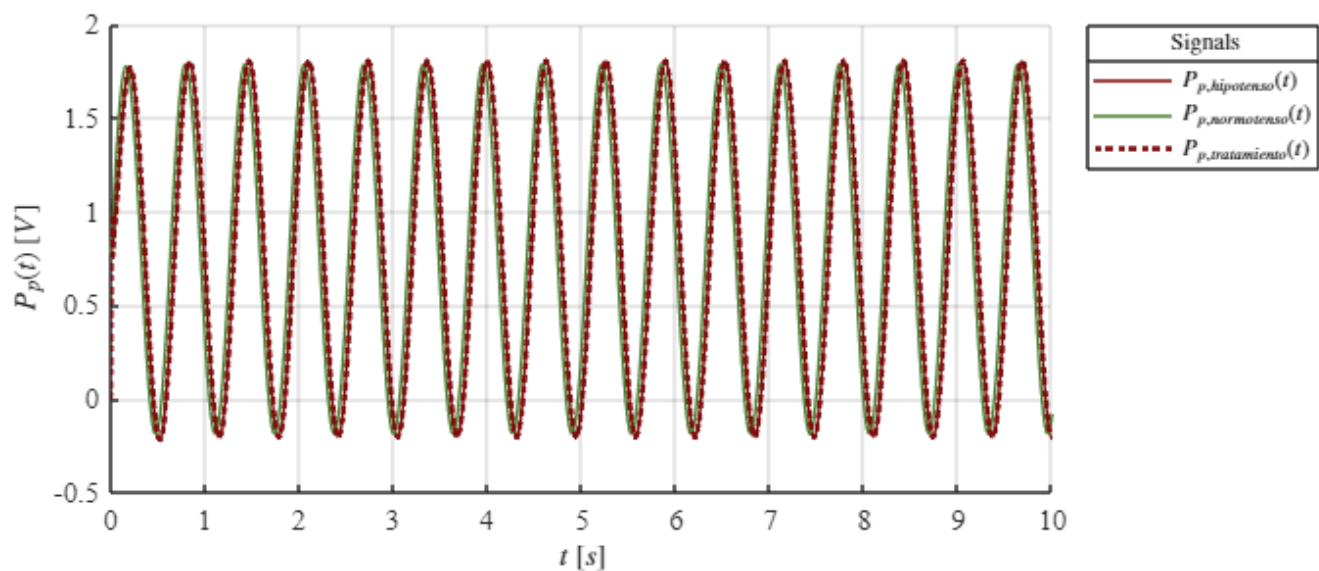
Respuesta la función sinusoidal

```
set_param('SistemaP3/Pa(t)', 'Amplitude', '1');
set_param('SistemaP3/Pa(t)', 'Frequency', '2*pi*95/60');
set_param('SistemaP3/Pa(t)', 'Bias', '0.8');
N = sim(file, parameters);
plotsignals(N.t, N.Ppx, N.Ppy, N.Ppz)
```



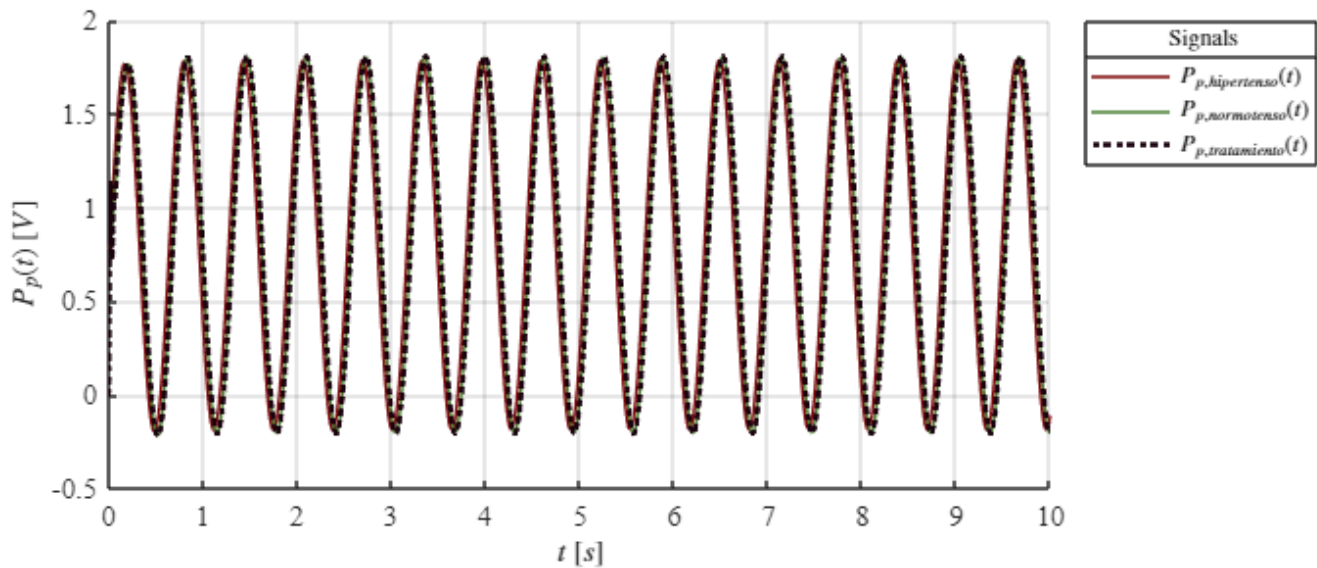
Hipotenso

```
clc; clear all; close all;
tend = '10';
file = 'Hipotenso';
open_system(file);
parametrer.StopTime = tend;
parameters.Solver = 'ode15s';
X = sim(file,parameters);
plothipotenso(X.t,X.Py,X.Px,X.Pz)
```



Hipertenso

```
clc; clear all; close all;
tend = '10';
file = 'Hipertenso';
open_system(file);
parametrer.StopTime = tend;
parameters.Solver = 'ode15s';
Y = sim(file,parameters);
plothipertenso(Y.t,Y.Px,Y.Py,Y.Pz)
```



Función: Respuesta a las señales

```
function plotsignals(t,Ppx,Ppy,Ppz)
set(gcf,'color','w')
set(gcf,'units','centimeters','position',[1,5,30,12])
set(gca,'FontName','Times New Roman','FontSize',11)
hold on; grid on

plot(t,Ppx,'LineWidth',0.5,'Color',[0,0.25,0.4])
plot(t,Ppy,'LineWidth',0.5,'Color',[0.3,0.5,0.2])
plot(t,Ppz,'LineWidth',0.5,'Color',[0.5,0.05,0.05])

L = legend('$P_{p,hipotenso}(t)$','$P_{p,normotenso}(t)$','$P_{p,hipertenso}(t)$');
set(L,'Interpreter','Latex','FontSize',9,'Location','NorthEastOutside')
title(L,'Signals')

xlabel('$t$ [s]','Interpreter','Latex','FontSize',11)
ylabel('$P_p(t)$ [V]','Interpreter','Latex','FontSize',11)

xlim([0,10]); xticks(0:1:10)
ylim([-0.5,2]); yticks(-0.5:0.5:2)
```

```

name = 'simulink_sistema.pdf';
exportgraphics(gcf,name,'ContentType','Vector')
end

```

Función: Tratamiento (Hipotenso)

```

function plothispotenso(t,Px,Py,Pz)
set(ffigure(),'color','w')
set(gcf,'units','centimeters','position',[1,5,30,12])
set(gca,'FontName','Times New Roman','FontSize',11)
hold on; grid on

plot(t,Px,'LineWidth',0.5,'Color',[0.5,0.05,0.05])
plot(t,Py,'LineWidth',0.5,'Color',[0.3,0.5,0.2])
plot(t,Pz,':','LineWidth',2,'Color',[0.5,0.05,0.05])

L = legend('$P_{p,hipotenso}(t)$','$P_{p,normotenso}(t)$','$P_{p,tratamiento}(t)$');
set(L,'Interpreter','Latex','FontSize',9,'Location','NorthEastOutside')
title(L,'Signals')

xlabel('$t$ $[s]$', 'Interpreter','Latex','FontSize',11)
ylabel('$P_p(t)$ $[V]$', 'Interpreter','Latex','FontSize',11)

xlim([0,10]); xticks(0:1:10)
ylim([-0.5,2]); yticks(-0.5:0.5:2)

name = 'simulink_hipotenso.pdf';
exportgraphics(gcf,name,'ContentType','Vector')
end

```

Función: Tratamiento (Hipertenso)

```

function plothispertenso(t,Px,Py,Pz)
set(ffigure(),'color','w')
set(gcf,'units','centimeters','position',[1,5,30,12])
set(gca,'FontName','Times New Roman','FontSize',11)
hold on; grid on

plot(t,Px,'LineWidth',0.5,'Color',[0.5,0.05,0.05])
plot(t,Py,'LineWidth',0.5,'Color',[0.3,0.5,0.2])
plot(t,Pz,':','LineWidth',2,'Color',[0.1,0.01,0.06])

L = legend('$P_{p,hipertenso}(t)$','$P_{p,normotenso}(t)$','$P_{p,tratamiento}(t)$');
set(L,'Interpreter','Latex','FontSize',9,'Location','NorthEastOutside')
title(L,'Signals')

```

```

xlabel('$t$ $[s]$', 'Interpreter', 'Latex', 'FontSize', 11)
ylabel('$P_p(t)$ $[V]$', 'Interpreter', 'Latex', 'FontSize', 11)

xlim([0,10]); xticks(0:1:10)
ylim([-0.5,2]); yticks(-0.5:0.5:2)

name = 'simulink_hipertenso.pdf';
exportgraphics(gcf,name,'ContentType','Vector')
end

```